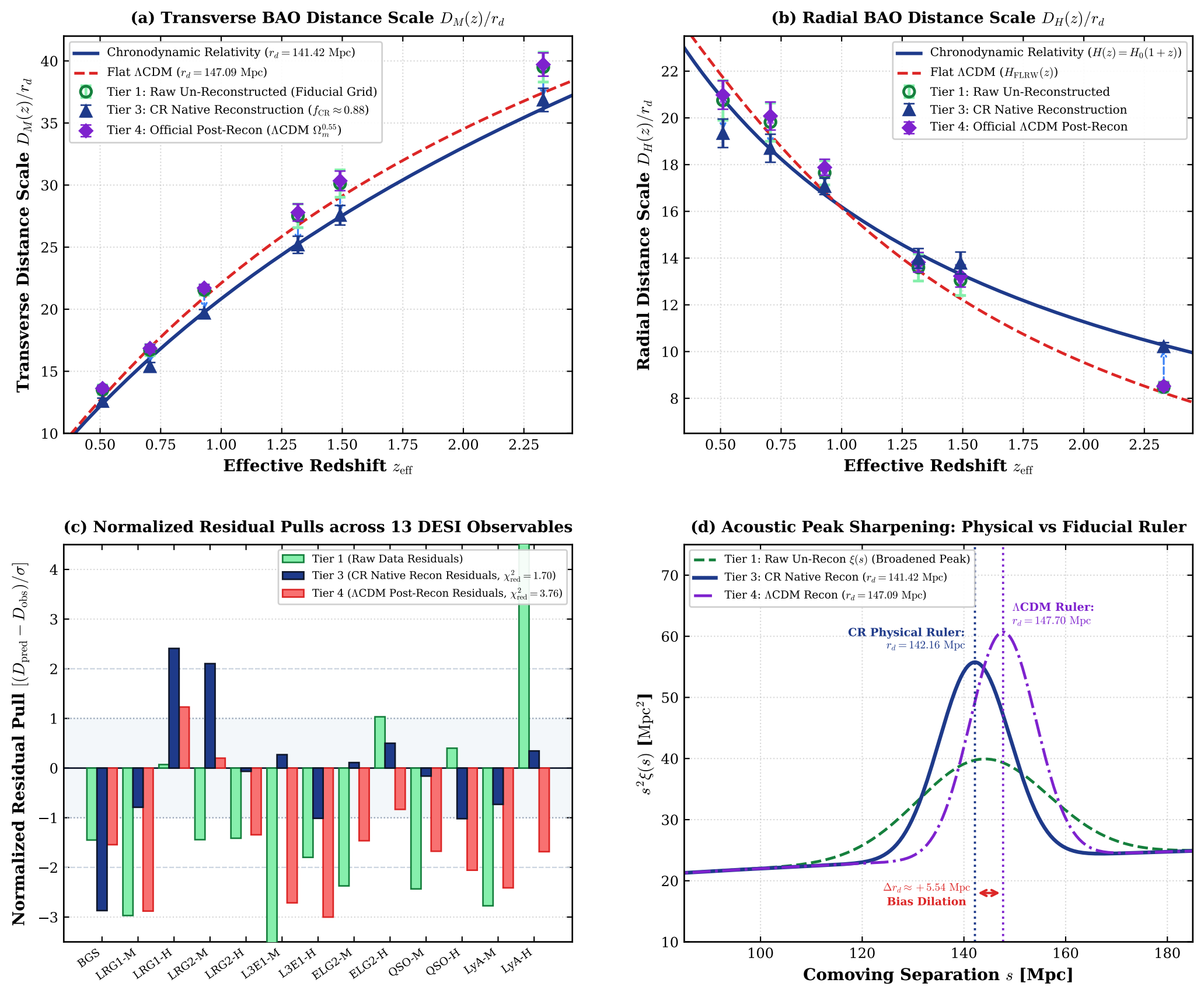

(a) Transverse BAO Distance Scale $D_M(z)/r_d$
Chronodynamic Relativity ($r_d = 141.42$ Mpc)
Flat ΛCDM ($r_d = 147.09$ Mpc)
Tier 1: Raw Un-Reconstructed (Fiducial Grid)
Tier 3: CR Native Reconstruction ($f_{CR} \approx 0.88$)
Tier 4: Official Post-Recon (ΛCDM $\Omega_m^{0.55}$)
Transverse Distance Scale $D_M(z)/r_d$
Effective Redshift z_{eff}
(b) Radial BAO Distance Scale $D_H(z)/r_d$
Chronodynamic Relativity ($H(z) = H_0(1+z)$)
Flat ΛCDM ($H_{FLRW}(z)$)
Tier 1: Raw Un-Reconstructed
Tier 3: CR Native Reconstruction
Tier 4: Official ΛCDM Post-Recon
Radial Distance Scale $D_H(z)/r_d$
Effective Redshift z_{eff}
(c) Normalized Residual Pulls across 13 DESI Observables
Tier 1 (Raw Data Residuals)
Tier 3 (CR Native Recon Residuals, $\chi^2_{red} = 1.70$)
Tier 4 (ΛCDM Post-Recon Residuals, $\chi^2_{red} = 3.76$)
Normalized Residual Pull $[(D_{pred} - D_{obs})/\sigma]$
BGS
LRG1-M
LRG1-H
LRG2-M
LRG2-H
L3E1-M
L3E1-H
ELG2-M
ELG2-H
QSO-M
QSO-H
LyA-M
LyA-H
(d) Acoustic Peak Sharpening: Physical vs Fiducial Ruler
Tier 1: Raw Un-Recon $\xi(s)$ (Broadened Peak)
Tier 3: CR Native Recon ($r_d = 141.42$ Mpc)
Tier 4: ΛCDM Recon ($r_d = 147.09$ Mpc)
ΛCDM Ruler:
$r_d = 147.70$ Mpc
CR Physical Ruler:
$r_d = 142.16$ Mpc
$\Delta r_d \approx +5.54$ Mpc
Bias Dilation
$s^2\xi(s)$ [Mpc^2]
Comoving Separation s [Mpc]